Developing a Non-Western Culture Aware RAG Application

Introduction: Most artificial intelligence (AI) models are trained predominantly on Western datasets, resulting in responses that often miss cultural nuances, misinterpret context, or provide inappropriate recommendations for non-Western users. Also, the language models are trained on English-heavy web data, which leads to weaker results for many non-Western languages and cultures. Multilingual benchmarks show strong English scores but clear drops elsewhere, indicating a systemic imbalance rather than isolated bugs [1]. Research in massively multilingual MT shows these gaps can shrink when models and data prioritize low-resource languages [2]. Recent research acknowledges cultural bias in AI systems [3, 4]. However, most solutions focus on post-hoc bias detection. They are not focused on considering the cultural knowledge while building the AI application. Our goal is to build an open-source production application that centers non-Western languages by developing a Retrieval-Augmented Generation (RAG) based web application that would be made freely available for everyone. RAG systems have demonstrated effectiveness in domain-specific applications [5, 6]. However, less work has been done on building RAG applications that are trained on non-Western culture data.

Purpose: Current large language models (LLMs) are not extensively trained on Western culture data. They are predominantly on Western datasets, resulting in responses that often miss cultural nuances, misinterpret context, or provide inappropriate recommendations for non-Western users. To overcome this difficulty, we propose to develop a Retrieval-Augmented Generation (RAG) based web application that addresses the significant cultural bias present in current AI applications.

Description: We will consider non-Western culture data that is open source [7]. We will extract the text and break it into smaller parts called tokens. We will then create a mathematical representation called an embedding for each token. These embeddings will be saved in a vector database. When a user inserts a query, the query will be broken down into tokens, and an embedding will be created, and this will be matched with the vector database. We will use cosine similarity to match them. The result from the model will now contain a response that will contain a non-Western cultural context. The developed application would be deployed using the open-source platform Hugging Face [8] and would be made freely available. The overview of our proposed RAG project is shown in Figure 1.